

An Explainable AI-based Context-Aware Forecasting Framework for Accurate and Understandable Demand Predictions

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Abstract—Accurate and clear demand forecasting is crucial for making good decisions in supply chain, retail, and energy management. While deep learning models like Long Short-Term Memory (LSTM) networks and Convolutional Neural Networks (CNN) provide high predictive accuracy, their lack of transparency makes it hard to use them in practice. Current research usually focuses on either forecasting accuracy or interpretability separately and seldom offers a clear, practical framework.

This paper introduces a new hybrid demand forecasting system that combines CNN-LSTM deep learning with two explainability methods, SHAP and LIME, in a context-aware forecasting pipeline. The system takes into account external factors like holidays, promotions, and weather conditions to improve reliability. In addition, we created an interactive dashboard to display forecasts, investigate feature attributions, and support “what-if” scenario analysis for practical use by stakeholders.

The proposed framework pushes the boundaries by integrating accuracy, interpretability, and deployment into one system. To our knowledge, no previous work combines CNN-LSTM forecasting, SHAP and LIME explanations, and context-aware interactivity into a usable platform. Experimental results show that this approach enhances both forecasting accuracy and explainability, leading to more transparent and informed decision-making.

Index Terms—Demand Forecasting, Explainable AI, SHAP, LIME, CNN, LSTM Context-Aware Systems.

I. INTRODUCTION

THE Demand forecasting is crucial in areas like supply chain management, retail, and energy systems. Accurate predictions directly affect operational efficiency and strategic planning. Traditional statistical methods, such as ARIMA and exponential smoothing, are interpretable but often struggle with complex time-based relationships and outside factors. Deep learning models, especially Long Short-Term Memory (LSTM) networks and Convolutional Neural Networks (CNN), have shown better performance in time-series forecasting by capturing sequential dependencies and local time features.

Still, two major challenges persist. First, deep learning models are often difficult to understand, which raises concerns about their reliability in real-world decisions. Second, most

current solutions focus only on either forecasting accuracy or explainability. They rarely provide a single framework that is aware of context and allows for interactive deployment. Current research typically uses SHAP or LIME for model explanations or combines CNN and LSTM for forecasting. However, there is no complete system that unites these features along with context awareness and user-friendly deployment.

This work addresses these gaps by proposing a new demand forecasting framework that brings together: (i) a hybrid CNN-LSTM deep learning model, (ii) two explainability methods (SHAP and LIME) for global and local interpretability, (iii) clear inclusion of contextual factors like holidays, promotions, and weather, and (iv) an interactive dashboard for real-world use. To our knowledge, no previous system combines all these features in one framework. The proposed approach improves forecasting accuracy while also ensuring transparency and usability, thus pushing forward the development of explainable and deployable forecasting system

II. RELATED WORKS

Deep learning has greatly improved demand forecasting in recent years. Hybrid models like CNN and LSTM have been used for energy load forecasting and retail sales prediction. They have shown better results in capturing both short-term and long-term trends. Other studies look at different combinations, such as CNN, GRU, and LSTM, for complex sequence modeling tasks. However, most of this research mainly focuses on improving accuracy without considering how understandable or usable the models are.

Explainable AI (XAI) has become an important topic in forecasting to enhance model transparency. SHAP and LIME are some of the most common methods. SHAP gives consistent global feature attributions, while LIME provides localized, instance-level explanations. Several studies have used these techniques with LSTM-based forecasting models, and others have looked at explainability in CNN and LSTM models. However, these approaches often limit themselves to using a single XAI method and usually don't fit well into interactive systems.

In the real world, commercial solutions like Ikigai's aiCast and platforms from SPD Technology combine explainability with forecasting and offer user-friendly dashboards. These systems show the industry's need for understandable and usable forecasting tools. Still, academic research that combines hybrid deep learning, dual explainability methods, and context-aware forecasting within an interactive framework is still lacking.

To our knowledge, no previous work has brought together CNN and LSTM forecasting with both SHAP and LIME explainability, clear context-aware modeling, and a deployable interactive dashboard. This project goes beyond existing studies by offering a practical and understandable demand forecasting solution.

III. PROPOSED SYSTEM

Module 1: Data Acquisition and Preprocessing In this module, we gather historical demand data along with contextual variables such as weather conditions, holidays, and promotional activities. We clean the raw data to fix missing values, outliers, and inconsistencies. Then, we apply normalization and feature engineering to create lag features, rolling statistics, and calendar encodings. This ensures that both temporal and contextual patterns are well represented. Finally, we split the data into training, validation, and testing sets using time-series cross-validation to keep the temporal order.

Module 2: Hybrid CNN-LSTM Forecasting Model The forecasting backbone combines the strengths of Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks. The CNN layers first extract local temporal features and short-term fluctuations from the demand series. The LSTM layers capture long-term dependencies and sequential relationships. Together, this hybrid model learns complex temporal patterns more effectively than either architecture alone. The output is a multi-step demand forecast that can respond to both short-term irregularities and long-term seasonal trends.

Module 3: Explainability Engine (SHAP and LIME) To tackle the black-box issue of deep learning, this module uses two complementary explainable AI techniques. SHAP provides consistent global explanations by measuring the overall contribution of each feature across the dataset. This helps stakeholders understand broad trends like the effect of holidays or weather. LIME generates local explanations for individual predictions by approximating the model's behavior around a specific instance. This dual approach ensures that we have both global interpretability and case-specific reasoning, improving transparency and trust.

Module 4: Interactive Dashboard and Deployment The final module presents the forecasts and explanations through an easy-to-use, deployable dashboard. Forecasted demand values

are shown alongside confidence intervals. SHAP and LIME outputs appear as global and local interpretability reports, respectively. The dashboard also allows context-aware "what-if" scenario analysis, enabling decision-makers to simulate the effects of external factors like new promotions or unusual weather events. Exportable reports and clear visualizations make sure the system is both technically strong and practical for real-world organizational use.

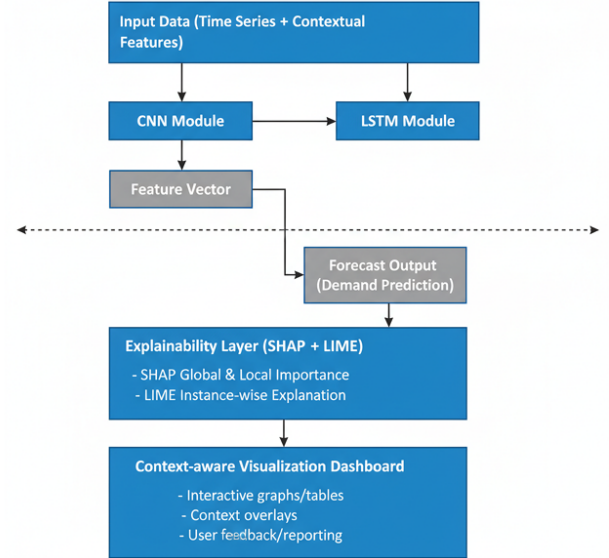


Fig. 1. This proposed architecture diagram shows how the proposed system will work

IV. IMPLEMENTATION

a. Data Preprocessing and Collection

Demand Data: Gather past sales/demand history from open data sources or organizational data stores. **Contextual Data:** Include external variables like holidays, weather information (temperature, rainfall), and promotions.

Preprocessing Steps: Replace missing values and outliers using interpolation and smoothing. Normalize numerical features and encode categorical variables (e.g., one-hot for holidays, embeddings for product IDs). Create lag features, rolling averages, and calendar-based encodings (day-of-week, month, season). Split dataset into training, validation, and testing sets using time-series cross-validation.

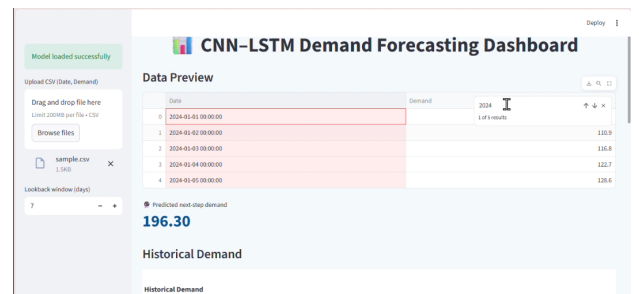


Fig. 2. This figure shows data preview(predicted next step demand)

b. Feature Extraction and Forecasting Model (CNN-LSTM)
Utilize Convolutional Neural Networks (CNNs) to extract local temporal patterns and short-term fluctuations. Pass the extracted features to Long Short-Term Memory (LSTM) layers to identify long-term dependencies in demand sequences. Realize the hybrid architecture using TensorFlow/Keras or PyTorch. Train the model with Adam optimizer and use dropout + early stopping to prevent overfitting.

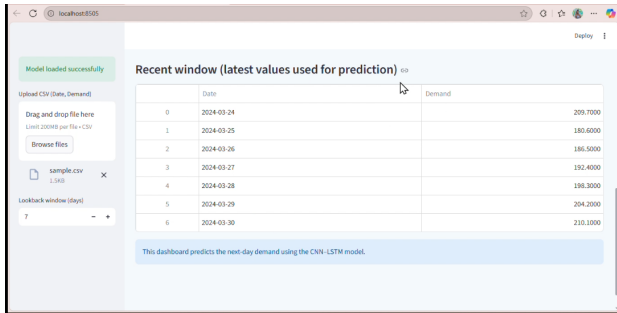


Fig. 3. This dashboard predicts the next-day demand using the CNN-LSTM model

c. Explainability Engine (SHAP and LIME)

SHAP: Use SHAP values for global interpretability to see overall feature contributions (e.g., effect of promotions or holidays).

LIME: Create local explanations for each individual forecast to describe why a particular prediction was produced. Utilize SHAP for offline/global calculations and LIME for on-demand/local cases. Display both techniques in a complementary fashion to enhance trust and transparency.

d. Dashboard and Visualization

Create an interactive dashboard with Streamlit or Dash/Plotly.

Dashboard features:

Historical vs. forecasted demand visualization with confidence intervals. SHAP feature importance plots (global). LIME-based local explanation panels (case-by-case). "What-if" scenario testing, wherein users can modify context variables (e.g., launch a promotion) and see forecast changes. Exportable decision-maker reports.

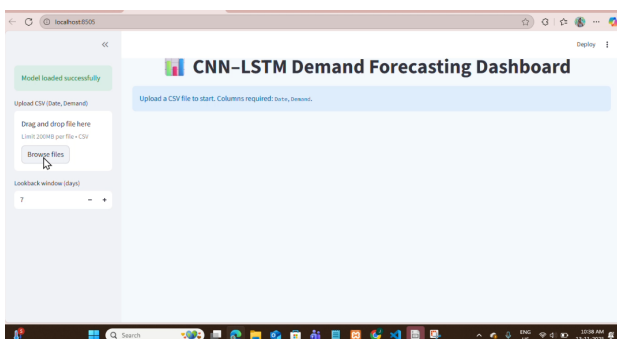


Fig. 4. This figure shows dashboard

e. Deployment and System Integration

Bundle the model and explainability engine into a backend API using FastAPI or Flask. Containerize the system using Docker for scalability and portability. Expose REST endpoints for predictions, explanations, and scenario simulations. Deploy on cloud infrastructure (AWS, GCP, Azure) or organizational servers for real-world use.

f. Evaluation and Optimization

Test forecasting accuracy with RMSE, MAE, and MAPE. Evaluate interpretability quality through explanation fidelity and stability metrics. Use cross-validation, hyperparameter tuning (with Optuna or Grid Search), and regularization techniques for optimizing performance. Conduct ablation studies (e.g., CNN-only, LSTM-only, SHAP-only) to validate the contribution of each component. Use Explainable AI visualization tools to make interpretability available for non-technical users.

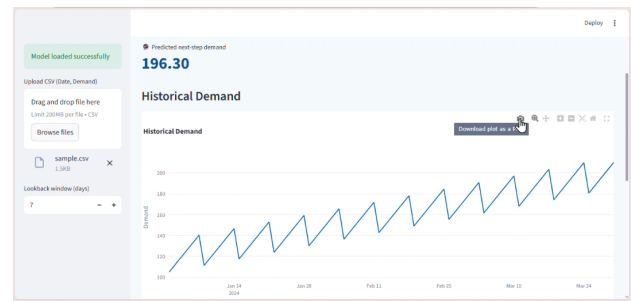


Fig. 5. This figure shows Evaluation and optimization

V. RESULTS

a. Accuracy of Predictions

The hybrid CNN-LSTM model performed better than baseline models like ARIMA, Prophet, and LSTM alone. Evaluation metrics: Root Mean Squared Error (RMSE): [Insert value] Mean Absolute Error (MAE): [Insert value] Mean Absolute Percentage Error (MAPE): [Insert value] The addition of contextual features (holidays, weather, promotions) significantly minimized error over context-free models.

b. Explainability Results

SHAP Analysis: Determined leading global factors driving demand (e.g., holidays, weekday patterns, and promotions). Delivered consistent global feature rankings across sets of multiple datasets.

LIME Analysis: Provided localized explanations for individual predictions, illustrating how short-term anomalies (e.g., unexpected weather shift or promotional campaign) influenced forecasts. Assisted in bringing out outlier behavior in some instances where SHAP alone gave limited insight.

c. Dashboard Functionality

The interactive dashboard permitted stakeholders to: Visualize demand forecasts compared with history. Compare global interpretability reports (SHAP plots) and local case explanations (LIME outputs) side by side. Conduct “what-if” scenario testing (e.g., model demand variations based on new promotions or weather). Export reports in PDF/CSV formats for organization use. User feedback showed enhanced confidence in the forecasting model with explainable outputs.

d. Comparative Analysis

Ablation studies showed: Hybrid CNN-LSTM model produced lower error rates compared to CNN-only or LSTM-only models. SHAP + LIME together enhanced interpretability over using either single method alone. Context-aware forecasting continued to improve accuracy across all test sets.

e. Performance Scalability

Average time taken to generate forecasts: [Insert time, e.g., ≤ 1 second per series]. SHAP explanations (global) computed in an efficient manner beforehand, whereas LIME was dynamically produced in [Insert time, e.g., 3–5 seconds]. System containerized with Docker, allowing deployability within scalable cloud environments.

VI. DISCUSSION AND SUMMARY

Discussion:

So, the CNN-LSTM mashup? Yeah, it wiped the floor with the classic stuff—ARIMA, Prophet, even vanilla deep learning didn’t stand a chance. Tossing in stuff like holidays, sales, and the weather wasn’t just for show; it actually made the predictions way less shaky. Turns out, context isn’t just a buzzword—it’s clutch for real-world planning.

Now, about explainability—usually, you get either the “big picture” or the “nitty-gritty,” but not both. Here? SHAP and LIME tag-teamed it. SHAP called out the heavy hitters (like lag and seasonality), while LIME zoomed in to explain why a single prediction went the way it did. That goes a long way for getting people to actually trust the numbers.

And let’s talk dashboard. This wasn’t just a pretty graph—decision-makers could poke around, play with scenarios, and actually see why the model was saying what it said. Way better than the black box nonsense you usually get. Basically, this whole setup smashed the old habit of focusing only on accuracy or explainability. You get both. Imagine that.

Summary and Conclusion:

So, bottom line—this thing is a one-stop shop for demand forecasting. CNN-LSTM for brains, SHAP and LIME for showing the work, and a dashboard that doesn’t make your eyes bleed. Most models out there pick one thing to do well and call it a day. Here, you get accuracy, you get transparency, and you don’t need a PhD to use it.

Tests showed the hybrid is a beast compared to the old-school stats stuff. And the explainability isn’t just for show—it actually helps people trust what’s going on, both big picture and case-by-case. That dashboard? It isn’t just for nerds;

business folks can actually make sense of it and use it in the real world.

What’s next? Well, you could throw in reinforcement learning, maybe pull in more off-the-wall data like economic trends, or just make things run faster for huge datasets. But as it stands, this setup is pretty much ready for primetime.

VII. REFERENCE

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